

STATEMENT OF WORK

IDIQ FOR RADIO COMMUNICATIONS SITE INFRASTRUCTURE

UNITED STATES GEOLOGICAL SURVEY (USGS) EARTHQUAKE SCIENCE CENTER NORTHERN CALIFORNIA SEISMIC NETWORK

I. INTRODUCTION:

The United States Geological Survey (USGS) Earthquake Science Center operates the Northern California Seismic Network (NCSN). The NCSN records ground motion from all earthquakes in northern and central California, computes seismological and engineering parameters for earthquakes automatically and accurately within minutes of their occurrence, and immediately delivers this information to the public and emergency response agencies.

To meet the continuous, low-latency data transmission requirement for the ShakeAlert earthquake early warning system and broader earthquake and volcano monitoring, the USGS and its partners use radio communications links to transmit data from 550+ seismic monitoring stations to a centralized data collection center. This is achieved using point-to-point radio networks in a “hub and spoke” topology, with remote stations relaying data to mountaintop relay sites that host microwave backhaul systems, including 11 primary backbone nodes and multiple secondary relays.

The USGS is seeking qualified contractor support to ensure the continued reliability and performance of its radio communications site infrastructure. The scope of work encompasses maintenance, repair, and upgrades including, but not limited to, environmental control systems; generator and solar power systems; power distribution systems; backup battery arrays; earth grounding; communications shelter and tower infrastructure; antenna installation and alignment; coaxial cable routing; and remediation of deficiencies identified through inspection processes.

II. SCOPE OF WORK:

- A. This is an “Indefinite Delivery, Indefinite Quantity” (IDIQ) contract. Multiple task orders will be issued against this contract. Each task order will enumerate the specific requirements for the scope of work at each worksite. The work will focus on maintaining, repairing, and/or upgrading radio communications site infrastructure and systems, including but not limited to:
1. HVAC system inspection, repair or replacement, and annual preventive maintenance.
 2. Generator and automatic transfer switch (ATS) inspection, repair or replacement, and annual preventive maintenance.
 3. Solar power system inspection, repair, and preventive maintenance.
 4. Upgrade backup battery arrays.
 5. Upgrade power distribution systems.

6. Inspect all earth grounded systems. Ensure that all metal surfaces inside and outside the communications shelters, the generators, the propane tanks, the solar arrays, and the towers are grounded in accordance with Motorola R-56 standards.
7. Construct new communications shelters.
8. Inspect communications shelter roofs, ceilings, walls, doors, and floors for damage, deterioration, and structural integrity. Repair or replace as needed.
9. Inspect communications shelter electrical panels and outlets. Repair as needed.
10. Inspect communications shelter fire extinguishers and smoke detectors. Repair or replace as needed.
11. Construct new towers.
12. Inspect towers and mounting hardware for cracks, corrosion, rust, or other damage. Repair or replace as needed.
13. Inspect tower, shelter, generator, and fuel tank foundations for cracks, settling, or other damage. Repair or replace as needed.
14. Inspect antenna and microwave dish mounting hardware. Replace as needed.
15. Inspect antennas and microwave dishes for damage. Replace as needed.
16. Run coaxial cables from antennas on towers to equipment in radio racks. Inspect existing coaxial cables for abrasions, bend, or other damage.
17. Remediate deficiencies identified through Department of the Interior (DOI) radio communications site inspections.

B. The contractor will submit individual bids on each assigned task order. The USGS contracting officer will review each bid. The contractor may request a site visit to properly develop a quote for each task order.

C. Contractor Qualifications

1. The contractor shall have extensive experience with remote radio communications site infrastructure maintenance, repairs, and upgrades.
2. The contractor shall hold all applicable national, state, and local licenses.

D. Compliance Requirements

1. All work performed under this contract shall fully comply with all governing authorities, laws, ordinances, regulations, codes, and standards, including, but not limited to:
 - a. National Codes: ANSI/EIA/TIA 569 – Commercial Building Standard for Telecommunications Pathways and Spaces.
 - b. Industry Standards: Motorola R-56 Standards and Guidelines for Communication Sites.

III. TOOLS AND MATERIALS

A. Each task order will specify the tools, materials, and equipment that the contractor shall provide.

- B. The contractor is responsible for the transport of Government-furnished materials to each worksite and shall retrieve the materials from the designated pickup location.
- C. In all cases, the USGS shall provide escorted access to each worksite and materials pickup location.

IV. PLACE OF PERFORMANCE

- A. The work shall be performed at multiple remote radio communications sites throughout northern and central California, each with different access rules and requirements. **Appendix A** lists most of the worksites but is not an exhaustive list of all the possible sites.
- B. The USGS shall coordinate access to all worksites as required.

V. PERIOD OF PERFORMANCE

- A. This is a five (5) year IDIQ contract.

VI. COMPLETION CRITERIA AND DELIVERABLES

- A. After a task order has been issued and accepted, the contractor shall provide a proposed work schedule before any work begins.
- B. After work has been completed, the contractor shall provide written documentation, block diagrams, and/or as-built drawings as required per the statement of work for each task order. All documentation shall include photographs.
- C. After work has been completed, the contractor shall ensure that all debris, trash, tools, and surplus materials have been removed from the worksite.
- D. Work is not considered complete until USGS personnel have performed a final walkthrough and accepted the work as “complete.”

VII. SPECIAL CONSIDERATIONS

- A. The contractor is NOT responsible for programming radios or any other devices that require software configuration unless explicitly stated in a “statement of work” document.
- B. The contractor shall follow all Federal, State, and local laws and site-specific rules.
- C. The contractor and/or subcontractor will be required to show proof of current tower climbing and tower safety training prior to performing work on towers or elevated surfaces.
- D. The contractor and/or subcontractor will be required to demonstrate proof of license prior to performing work in any of the licensed trades. Licensed trades in California include general engineering and building contractors, as well as specialty trades such as electrical, plumbing,

HVAC, roofing, concrete, drywall, flooring, glazing, landscaping, masonry, fire protection, low-voltage systems, sheet metal, refrigeration, and many more under the CSLB classifications.

- E. The contractor shall apprise USGS personnel of any schedule changes. The contractor shall obtain written approval from USGS personnel before accessing a worksite on a non-scheduled day.
- F. If at any time while working on the site the contractor digs up or discovers any sensitive objects, work must stop immediately and the contractor must contact USGS personnel. Sensitive objects include, but are not limited to, bones, artifacts, historical items, buried pipes or wires, and other similar items.
- G. BE FIRE CONSCIOUS. Wildfires can start at any time of year. The Contractor must have at least two (2) fire extinguishers available and ensure all crew members know how to use them.

APPENDIX A

Table of possible worksites. Other sites may be added to Task Orders on this IDIQ.

NCSN Region	Site Name	County (CA)	Latitude	Longitude
Bear Valley	Palo Escrito	Monterey	36.40591	-121.49165
Calaveras	Bollinger Ridge	Contra Costa	37.81647	-122.06294
Calaveras	Mount Hamilton	Santa Clara	37.34693	-121.63003
Calaveras	Vollmer Peak	Contra Costa	37.88400	-122.22322
Calaveras	Walpert Ridge	Alameda	37.65182	-122.00203
Geysers	Geyser Peak	Sonoma	38.76469	-122.84518
Hollister	School Peak	Monterey	36.76013	-121.49122
San Jose	Loma Prieta	Santa Clara	37.11075	-121.84408
San Jose	Moffett Field	Santa Clara	37.42046	-122.05767
Mammoth	Mammoth Water District	Mono	37.64162	-118.94568
Napa	Sonoma Mountain	Sonoma	38.34882	-122.57793
Napa	Mount Tamalpais	Marin	37.92371	-122.59648
Parkfield	Black Mountain	San Luis Obispo	35.39445	-120.353898
Parkfield	Carr Hill	San Luis Obispo	35.88813	-120.43051
Parkfield	Hog Canyon	San Luis Obispo	35.86661	-120.47960
Parkfield	Tassajera Creek	Monterey	35.39269	-120.70787
Parkfield	Williams Hill	Monterey	35.95157	-121.00137